CONSTRUCTING CALCULATOR FOR BINARY ADDITION AND SUBTRACTION

LAB 5

CSE-202L DIGITAL LOGIC GATES

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ADDERS:

HALF ADDER

A Half Adder is a combinational circuit that performs the addition of two bits.

- We have two input variables and two output variables.
- The two output variables are called Sum (S), and Carry (C).
- The Sum variable stores the least significant bit while Carry stores the most significant bit of the output.

TRUTH TABLE

A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

DERIVING SUM FUNCTION

- Mapping Sum Function On K-Maps from Truth Tabe:

	B'	**B**
A'	0	1
A	1	0

USING MIN-TERMS:

$$F=m1+m3$$

$$F=A'B+AB'$$

$$F=A\oplus B$$

DERIVING CARRY FUNCTION

- Mapping Carry Function On K-Maps:

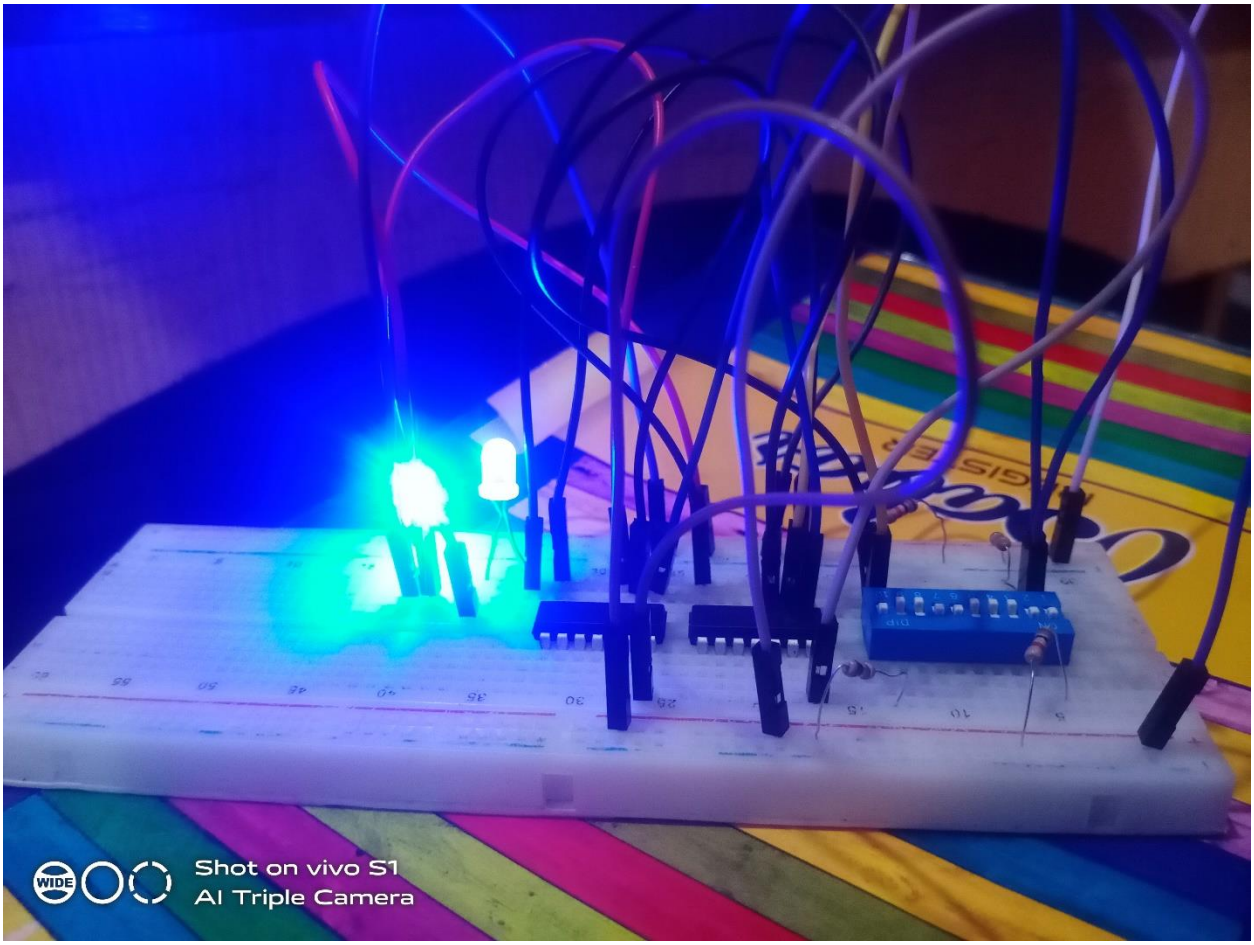
	B'	**B**
A'	0	0
A	0	1

USING MIN-TERMS:

F=m3

F=AB

LAB PERFORMANCE:



TRUTH TABLE DRAWN ON EXPERIMENTAL DATA:

A	B	C	S
L	L	L	L
L	H	L	H
H	L	L	H
H	H	H	L

FULL-ADDER

A Combinational circuit that performs the addition of two, one-bit numbers is called a Full-Adder. The addition is performed between the two, one-bit numbers and previous carry Cin.

- We have three input variables and two output variables.
- The three input variables include; input bit A, B, and previous carry Cin.
- The output variables are Sum(S), and Carry(C).
- Full adder can be implemented using two half adders.

TRUTH TABLE:

A	B	Cin	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

DERIVING SUM FUNCTION:

- Mapping Sum Function On K-Maps:

	B'Cin'	**B'Cin**	**BCin**	**BCin'**
A'	0	1	0	1
A	1	0	1	0

USING MIN-TERMS:

$$F=m1+m2+m4+m7$$

$$F=A'B'Cin+A'BCin'+AB'Cin'+ABCin$$

$$F=Cin(A'B'+AB) + Cin'(A'B+AB')$$

Let $A \oplus B=X$,

$$F=CinX'+Cin'X$$

$$F=Cin \oplus X$$

$$F=Cin\oplus(A\oplus B)$$

DERIVING CARRY FUNCTION:

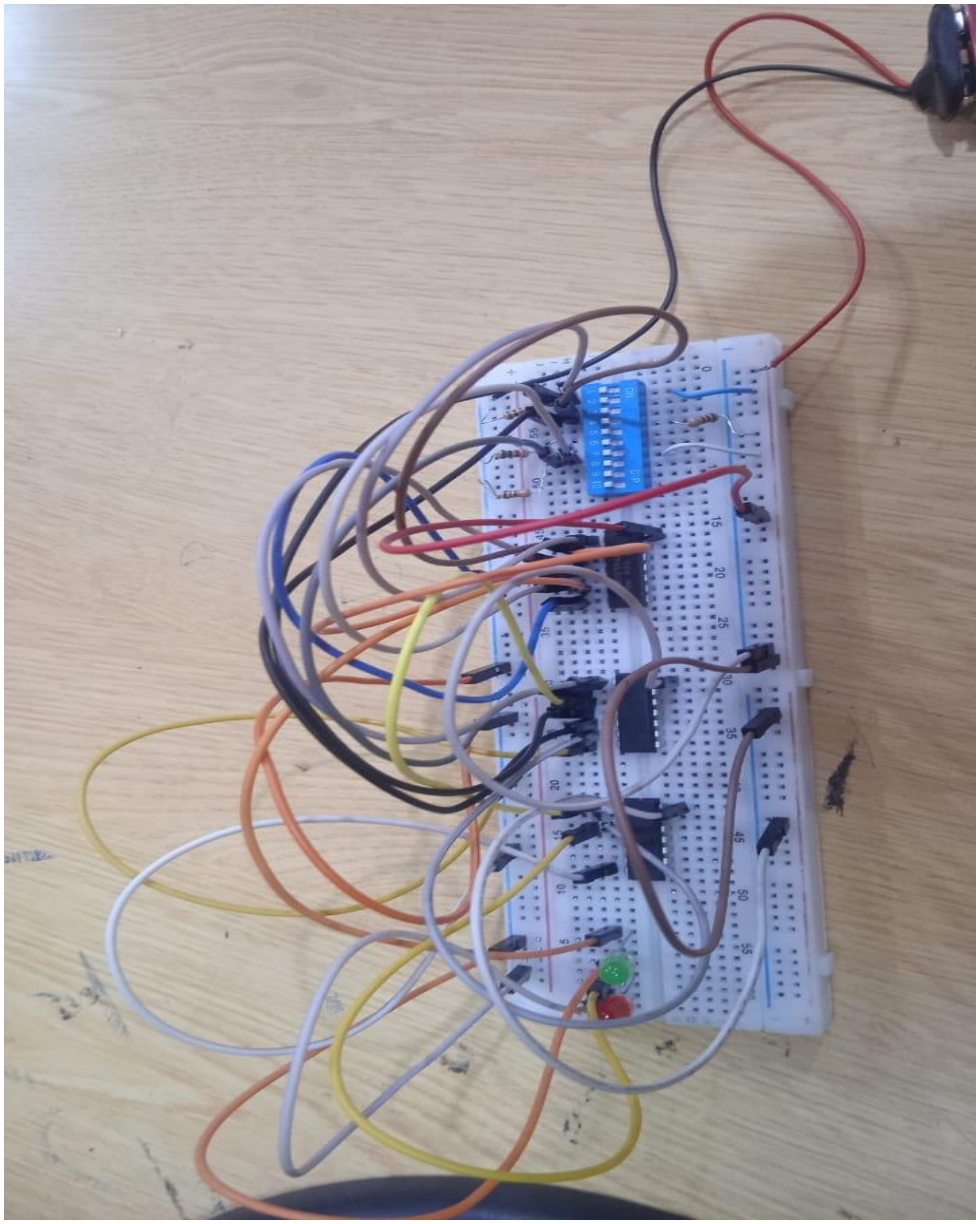
- Mapping Carry function on K-Maps:

	B'Cin'	**B'Cin**	**BCin**	**BCin'**
A'	0	0	1	0
A	0	1	1	1

USING MIN-TERMS:

$$F=m3+m5+m6+m7$$
$$F=A'BCin+AB'Cin+ABCin'+ABCin$$
$$F=Cin(A'B+AB')+AB(Cin'+Cin)$$
$$F=(Cin(A\oplus B))+AB$$

LAB PERFORMANCE:



TRUTH TABLE DRAWN ON EXPERIMENTAL DATA:

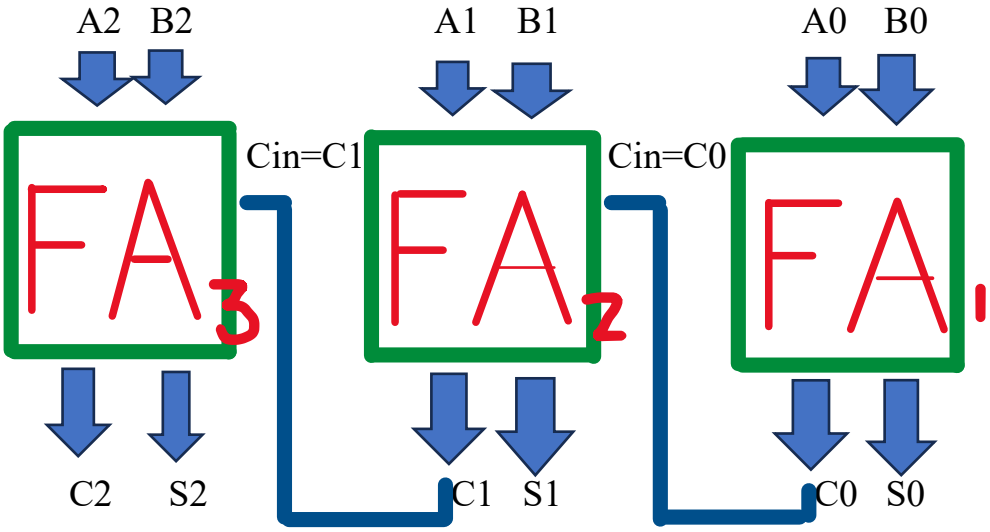
A	B	Cin	C	S
L	L	L	L	L
L	L	H	L	H
L	H	L	L	H
L	H	H	H	L
H	L	L	L	H
H	L	H	H	L
H	H	L	H	L
H	H	H	H	H

RIPPLE CARRY ADDER:

Half or Full-Adders are useless on their own because of their limit to add only two one-bit numbers. They are not instrumental to use in hardware. The more pragmatic approach is to cascade many full-adders to make a flexible one capable of adding more than one-bit numbers. Such an adder is known as a Ripple Carry Adder.

WORKING:

- Suppose, we want to add two, three-bit numbers, A2A1A0, and B2B1B0.
- We will implement the ripple carry adder by cascading 3 full-adders. Each adder will add two bits only, and pass any carry to the next one.
- The previous carry variable Cin for the first full-adder is equal to zero. Therefore, we can also replace the full adder with a half-adder.
- The first full adder adds A0, and B0. The second adder adds A1, and B1. The third adder adds A2, and B2.
- The least significant carry out C0 is passed as Cin to the second-full adder. Similarly, the carry out C1 from the second full adder is passed to the third one as Cin. Thus, resembling a ripple from where the adder gets its name.
- The Sum variable will compromise of three bits S2S1S0. In result, carry out C2 will be our most significant bit followed by the sum variable bits (C2S2S1S0).
- The output of the Sum and Carry OUT C2 is passed to LED.



IC: 7483 has 4 cascaded full-adders, and can add two 4-bit binary numbers.

SUBTRACTORS:

HALF-SUBTRACTOR

A Combinational circuit that performs the subtraction of two bits.

- We have two input variables A, B and two output variables D and Bout.
- D stands for difference and stores the difference of input bits.
- Bout stands for borrow out, and it stores values that are borrowed during subtraction of the bits.

TRUTH TABLE:

A	B	D	Bout
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

DERIVING DIFFERENCE FUNCTION:

	B'	**B**
A'	0	1
A	1	0

Using min-terms:

$$F=A'B+AB'$$

$$F=A\oplus B$$

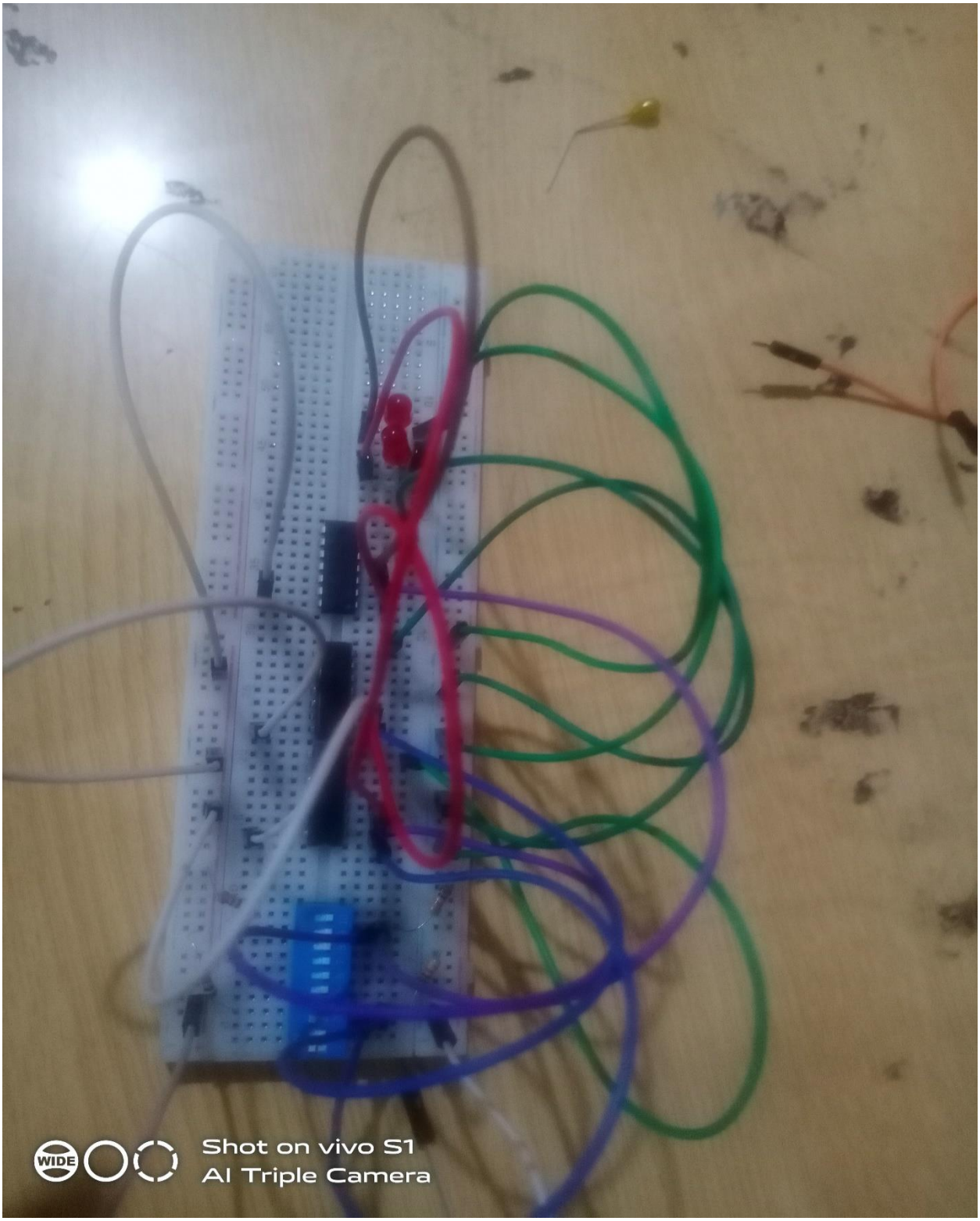
DERIVING BORROW FUNCTION:

	B'	**B**
A'	0	1
A	0	0

Using min-terms:

$$F=A'B$$

LAB PERFORMANCE:



TRUTH TABLE DRAWN ON EXPERIMENTAL DATA:

A	B	D	Bout
L	L	L	L
L	H	H	H
H	L	H	L
H	H	L	L

FULL SUBTRACTOR:

A combinational circuit that performs the subtraction of two 1-bit numbers. The difference between half and full subtractor is same just like in adders. We have a third input variable.

- In a full subtractor, there are 3 input variables and 2 output variables.
- The input variables are A, B and Bin. Where Bin is the input borrow which denotes the borrow that the bit under operation assumes.
- The two output variables are D (difference) and Bout (borrow out).

TRUTH TABLE:

A	B	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

DERIVING DIFFERENCE FUNCTION:

Mapping D on K-Maps:

	B'Bin'	**B'Bin**	**BBin**	**BBin'**
A'	0	1	0	1
A	1	0	1	0

Using min-terms:

F=A'B'Bin+A'BBin'+AB'Bin'+ABBin

$$F=Bin(A'B'+AB)+Bin'(A'B+AB')$$

$$F=BinX'+Bin'X$$

$$F=Bin\oplus X$$

$$F=Bin\oplus A\oplus B$$

DERIVING BORROW OUT FUNCTION:

Mapping Bout on K-Maps:

	B'Bin'	**B'Bin**	**BBin**	**BBin'**
A'	0	1	1	1
A	0	0	1	0

Using min-terms:

$$F=A'B'Bin+A'BBin+A'BBin+A'BBin'+A'BBin+ABBin$$

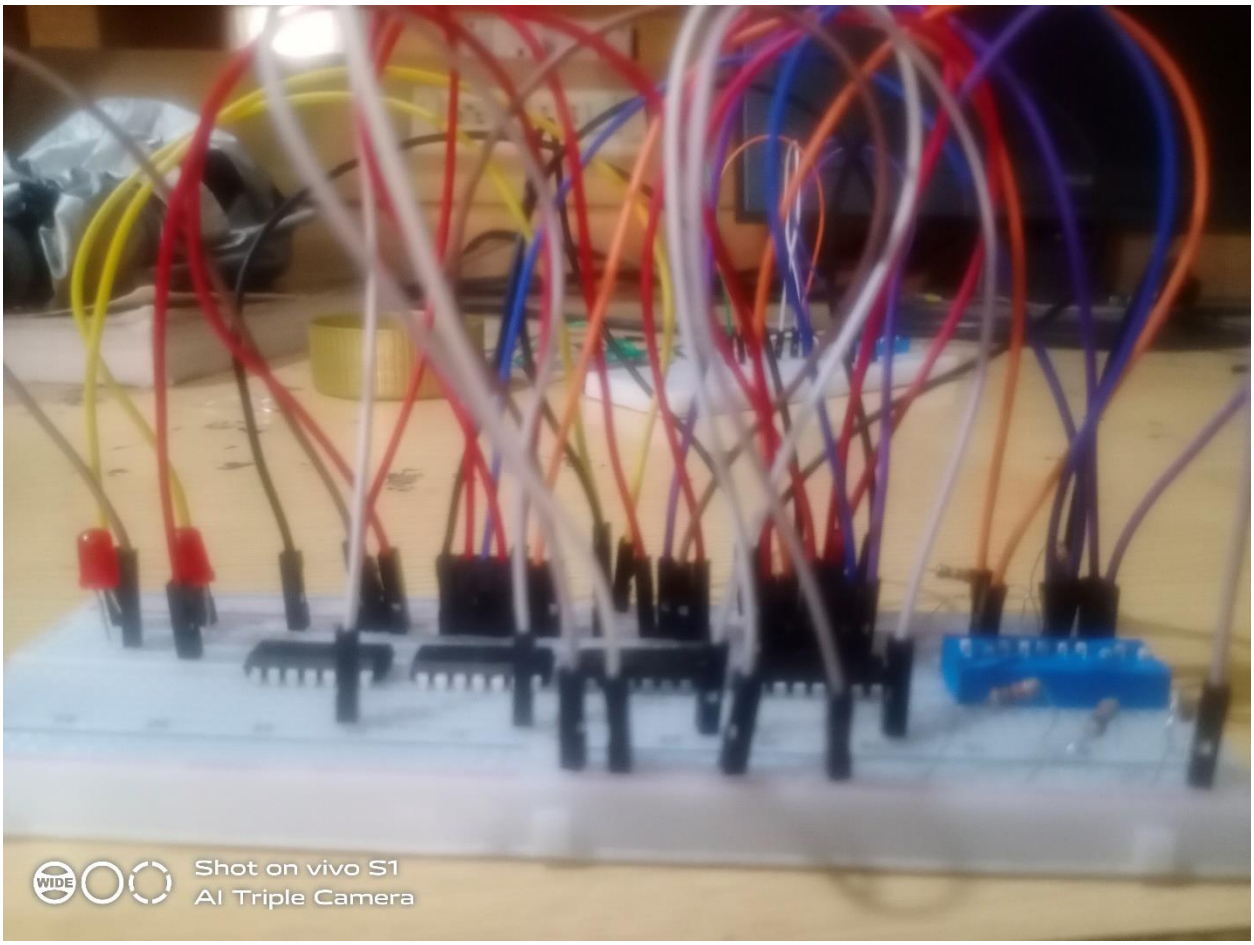
$$F=A'Bin(B'+B)+A'B(Bin+Bin')+BBin(A'+A)$$

$$F=A'Bin+A'B+BBin$$

$$F=Bin(A'B'+AB)+A'B(Bin+Bin')$$

$$F=Bin(A\odot B)+A'B$$

LAB PERFORMANCE:



TRUTH TABLE DRAWN ON EXPERIMENTAL DRAWN:

A	B	Bin	D	Bout
L	L	L	L	L
L	L	H	H	H
L	H	L	H	H
L	H	H	L	H
H	L	L	H	L
H	L	H	L	L
H	H	L	L	L
H	H	H	H	H

BINARY ADDER AND SUBTRACTOR USING 7483:

We can use two 7483 ICs along with two XOR ICs to implement an 8-bit binary adder. In this construction, subtraction is done via addition.

COMPONENTS:

2 7486 IC's

2 7483 IC's

1 DIP Switch

17, 1k Resistors

Jumper wires

CONNECTIONS:

For a 4-bit adder/subtractor, the connections are explained below:

- The first switch is our control switch. Through this control switch we will perform addition by turning the switch off, and subtraction by turning the switch on.
- The next 4 switches (from2-5) are the 4-bits of our first input, A3A2A1A0.
- From switch 6-9, we have passed the 4-bits of our second input, B3B2B1B0 to the four gates of the 7486 IC.
- We will now pass the first input directly from the switch to the 7483 adder IC. And the second input from the 4 outputs of the 7486 IC.
- Make connections following the pinout of the 7483 IC.
- Take result S3S2S1S0 from the corresponding pins and pass to LED's.
- Pass carry-out from pin 14, this will be our most significant bit of the result.

For an 8-bit adder/subtractor repeat the connections explained above. Pass carry-out from pin 14 of the first IC to the carry-in pin 13 of the second 7483 IC. The carry-out of second IC will be our most-significant bit of the result followed by S7S6S5S4S3S2S1S0.

While performing subtraction we have two terms,

$$\begin{array}{rl} 3 & \text{Minuend} \\ -\ 2 & \text{Subtrahend} \\ \hline \end{array}$$

SUBTRACTION VIA 2's COMPLEMENT:

- Add minuend with 2's complement of subtrahend.
- If there's an end-carry, ignore it.
- If there's no end-carry, take complement of the result and put a negative sign in front of it.

IMPLEMENTING 2's COMPLEMENT IN THE CIRCUIT:

- We are implementing the procedure for 2's complement by passing the 4-bits of given subtrahend to 7486 IC.
- Each of the four gates in the 7486 IC will take one bit of the subtrahend.
- The second input to all of the four gates of the IC will be passed from the control switch.
- Now we can perform addition or subtraction by turning the control switch ON and OFF.
- Addition is executed when the control switch passes an input of 0 to the 7486 IC.
- This zero-input passed to all four gates of the 7486 IC will not complement the individual bits. This is because whatever the other bit is, we know when zero is given as second input, the output is equal to the first bit.
- Now, to perform subtraction we will turn ON the control switch. When this control is passed to all four gates of the 7486 IC. It will complement each bit. This is because when the second input is 1, the first bit is complemented.

4-BIT ADDER/SUBTRACTOR

Addition performed for,

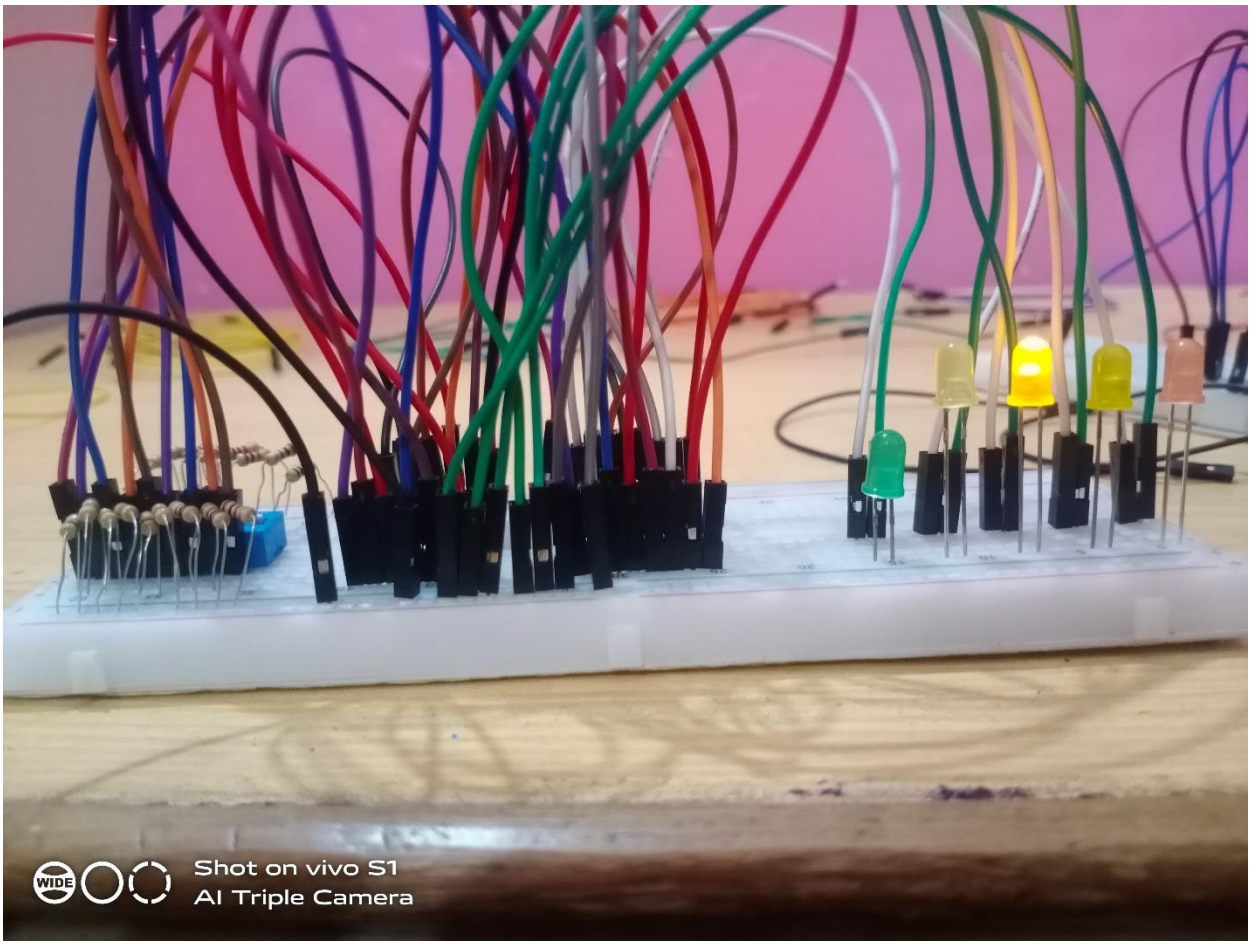
0001

0011

Output:0100

Leftmost led (green) is carry out, 0 in this case.

The remaining four LEDSs are S3S2S1S0. S2 is ON.



CONCLUSION:
In this lab we learned how a computer actually performs binary addition, and subtraction.